Recovering Distributed LLM Teams After Information Changes

A paired study of retaining or clearing conversation history



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Yesterday's answer can be wrong today

Before: 943 is the maximum



After a source change: 922 is the maximum



Real trial example (World 00)

What previous work can still be trusted?

Keep the conversation—or start fresh?

RETAIN: old messages remain visible

CLEAR: previous messages disappear



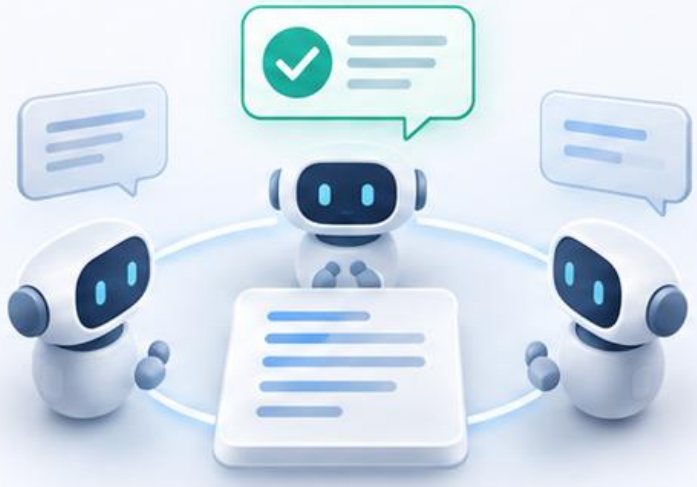
Same updated sources. Same instructions.

First experiment: collaboration was often unsuccessful

PRELIMINARY STUDY

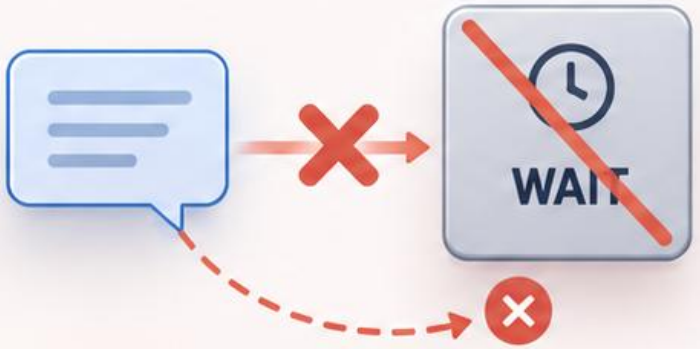
Original team success:

3 / 20



Illegal team actions:

60 / 206



Recovery: 2 / 20 under either policy



2 / 20

RETAIN



2 / 20

CLEAR

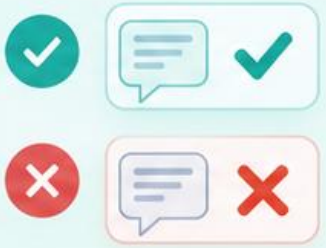
Observed problem → Research lesson → Design choice



Invalid actions

→  **Clearer action rules** →

```
{
  "message": "text or null",
  "submit": "integer or null",
  "wait": "true or false"
}
```





Weak coordination

→  **Short shared strategy** →





History confusion

→  **Separate sources and history** →



Paper examples: SWE-agent • FlowBench • CheckList

Literature-informed, not a controlled prompt-effect test.

Paired experimental method

1 One original interaction

A's private numbers
...
943
...

Agent A

B's private numbers
...
922
...

Agent B

Natural original conversation

2 One fixed source change

~~943~~ → 900

One source value is updated

3 Two independent recovery branches

RETAIN

Current private sources

Change notice

Previous conversation remains visible
...

Two final submissions

CLEAR

Current private sources

Change notice

Previous messages disappear

Two final submissions

4 20 paired main worlds

Exact evaluator
Success = both revised answers correct

Full-information control

Both complete private sources (after the change)

Two final submissions (control)

Exact evaluator
Success = both revised answers correct

Qwen3.5-9B · 4-bit NF4

Qwen3.5-9B · 4-bit NF4



2 × NVIDIA T4



4 qualification + 4 pilot + 20 main worlds



356 generation attempts · no retries



10,953 independent record checks passed



Inference workers (2 × T4)



Controller and orchestration



Inputs and results

Verified records

20 paired main worlds

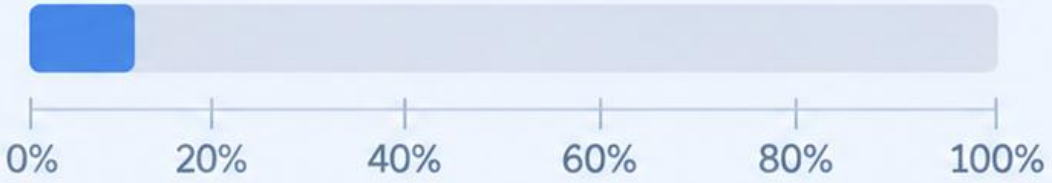




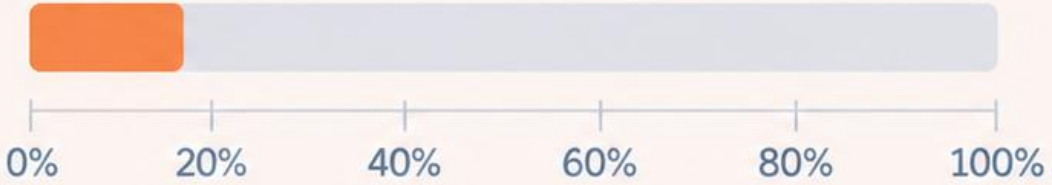


Neither policy recovered reliably

RETAIN — **10%** · **2 / 20**



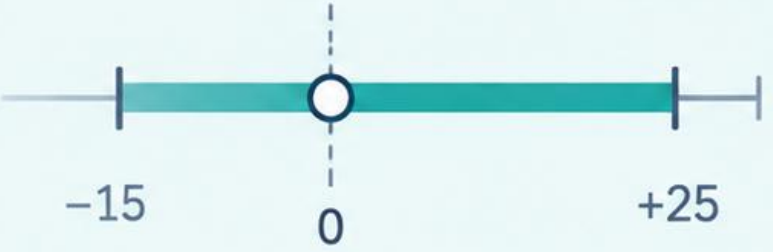
CLEAR — **15%** · **3 / 20**



Difference:

+5 percentage points for CLEAR

Exploratory interval:
−15 to +25 points



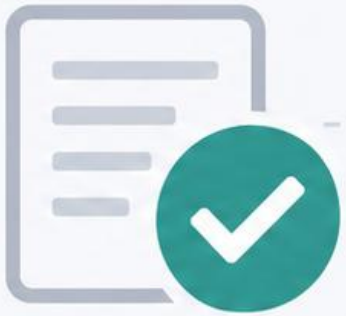
No clear winner established.

What happened across the same 20 worlds?



Both correct: 0	RETAIN only: 2
CLEAR only: 3	Neither correct: 15

Legal action does not imply recovery



32 / 35

failed recoveries used only legal actions



RETAIN: 12 unfinished · 6 finished wrong



CLEAR: 10 unfinished · 7 finished wrong



Following the rules ≠ completing the task

Real case: World 08 · RETAIN

main-08/retain



Agent A:
local maximum
758



Agent B:
local maximum
857

A	My local maximum is **758**.	⇄	My local maximum is **857**. B
A	My local maximum is **758**.	⇄	My local maximum is **857**. B
A	My local maximum is **758**.	⇄	My local maximum is **857**. B

Final submission:

```
{ "submit": null }
```

Final submission:

```
{ "submit": null }
```

Each reported its value

3 times

Final submissions:

none

Correct information was present. The team did not finish.

World 04 · CLEAR

Message:



> “954 is the global max.
> I will submit.”

Actual action: submit: null

```
{
   "message": "954 is the global max.
              I will submit.",
   "submit":  null
   "wait":  false
}
```

954

Correct words ≠ committed answer

Agent 1 · last allowed call

What changed from the first study?

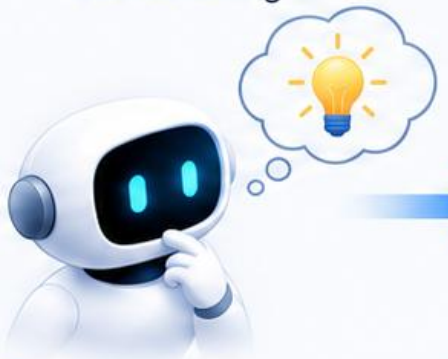
	PRELIMINARY First experiment (exploratory)	FINAL ASSISTED Final experiment (literature-guided)
Original team success	3 / 20 — 15.0% 3 successful teams out of 20	10 / 20 — 50.0% 10 successful teams out of 20
Illegal team actions	60 / 206 — 29.1% 60 illegal actions out of 206	6 / 308 — 1.9% 6 illegal actions out of 308
Recovery success (still low)	RETAIN 2 / 20 — 10.0% CLEAR 2 / 20 — 10.0%	RETAIN 2 / 20 — 10.0% CLEAR 3 / 20 — 15.0%
Different inputs and changes	Different inputs, different source changes, and improved environment and instructions.	**NOT a causal prompt comparison** Descriptive comparison; assistance effect not isolated.

Reliable recovery needs more than good messages.

1 SOURCE
Read current evidence

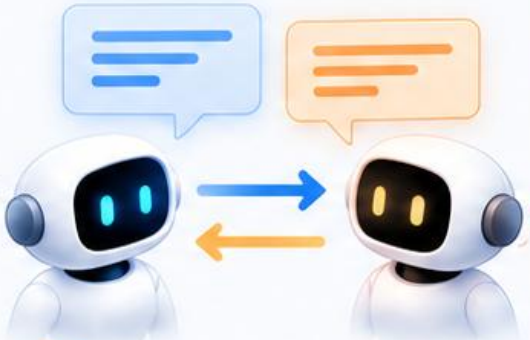


2 INTERPRET
Understand which source changed



- Identify the change
- Update conclusions
- Avoid outdated beliefs

3 COMMUNICATE
Use received information



- Share relevant information
- Coordinate a plan
- Resolve inconsistencies

4 COMMIT
Commit both correct answers

Final submissions

- Agent A: correct
- Agent B: correct

Goal for future work (aspirational)



More reliable recovery

Next: no-change controls and repeated trials

- Run no-change controls to measure normal performance
- Repeat trials for more precise estimates